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Structural Reinterpretation of the Ajo Mining District, Pima County, Arizona, Based on Paleomagnetic and Geochronologic Studies

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Abstract

The Ajo mining district of southern Arizona is divided into two main structural blocks by the Gibson Arroyo fault. The eastern Camelback Mountain block contains the Late Cretaceous-early Tertiary porphyry copper deposit which has been previously thought to be associated with the displaced apex of a large intrusion exposed by deeper erosion in the western Cardigan Peak block. However, unpublished U-Pb data support a mid-Tertiary age for the western intrusion. Paleomagnetic and geologic evidence indicates that the ore deposit has been tilted to the south a total of approximately 120°: 68° before and another 55° after emplacement of the overlying Locomotive Fanglomerate and Ajo Volcanics. The Ajo Volcanics have K-Ar ages (24–25 m.y.) slightly older than those from the large intrusion of the Cardigan Peak block (20–23 m.y.). Paleomagnetic directions are consistent with the southward tilting of the volcanic rocks, but they suggest that the remanent magnetization and perhaps the K-Ar dates of the western intrusion were reset by the emplacement of dikes younger than the Locomotive Fanglomerate and Ajo Volcanics. These and other geologic relations indicate the following sequence of mid-Tertiary events in the district: (1) emplacement of the western intrusion, (2) movement along the Gibson Arroyo fault, (3) unroofing and perhaps tilting of the pluton $\simeq$70° to the south along with the Camelback Mountain block, (4) syntectonic depositions of the Locomotive Fanglomerate and the Ajo Volcanics, (5) continued uplift and tilting to the south totaling 40° to 60°, (6) intrusion of the youngest dikes with attendant alteration and remagnetization of the host rocks, and (7) minor (?) oblique movement along the Gibson Arroyo fault.

Introduction

THE Ajo mining district is part of the porphyry copper province centered in southeastern Arizona. Located in the Little Ajo Mountains of western Pima county (Fig. 1), the deposit at Ajo is typical of the province in its association with a Late Cretaceous-early Tertiary (Laramide) intrusion. The intrusion is relatively isolated from other mineralized plutons in the region and lies within a wide area of mid-Tertiary volcanics. The intrusion at Ajo is hosted by a thick sequence of volcanic rocks, and the associated orebody is similar to the porphyry copper-gold-type deposit of the Lone Star mining district in Graham county, Arizona, described by Langton and Williams (1982). One of the first copper districts to be mined by Americans after the Gadsden Purchase of 1853, the deposit at Ajo has produced approximately 2.5 million tons of copper, with gold and molybdenum by-products, between 1917 and 1984 when production was suspended.

The first comprehensive study of the geologic framework of the Ajo mining district was that of Gilluly (1946). He concluded that the orebody is associated with the displaced apex of a large pluton exposed by deeper erosion to the west. Detailed grid mapping of this large pluton by Wadsworth (1968) made possible its subdivision into multiple intrusive phases and showed petrologic similarities between a portion of the pluton and the ore zone, supporting Gilluly's interpretation. K-Ar ages of 63 to 65 m.y. (McDowell, 1971) confirmed the Laramide age of the mineralized rocks but indicated an age of approximately 20 m.y. for the pluton to the west (Table 1). This discrepancy was ascribed by McDowell (1971) to a younger episode of alteration in the western pluton. Cox et al. (1981) described highly saline fluid inclusions in igneous quartz grains from the western pluton which also contain chalcopyrite daughter minerals. This reinforced the hypothesis that hydrothermal fluids had evolved from the crystallizing pluton and formed the porphyry copper deposit now faulted down to the east. Cox and Ohta (1984) remapped the pluton, revising Wadsworth's (1968) work, and noted sodic-calcic alteration assemblages like those recognized in the deep zones of the Yerington porphyry copper deposit in Nevada (Carten, 1979). This observation also supported a cogenetic relationship between the pluton and the orebody. However, this relationship recently has been put in doubt by unpublished U-Pb data (L. T. Silver, oral commun., 1983; J. S. Stacey and J. L. Wooden, written commun., 1986; R. M. Tosdal, written commun., 1987) that apparently corroborate McDowell's ages in suggesting a mid-Tertiary age for some rocks in the western pluton.

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TABLE 1. K-Ar Ages of Samples from the Ajo Mining District

Sample number	Rock type	Material analyzed	K_2O (%)	$^{40}Ar^*$ (moles/g)	$^{40}Ar^*/^{40}Ar$	Apparent age (m.y.)	Reference
Camelback Mountain block (New Cornelia mine)							
L-843	Granodiorite (?)	Biotite	3.49	3.96×10^{-10}	82	64.2 ± 1.9	1
				4.04×10^{-10}	78	65.2 ± 1.9	1
				3.94×10^{-10}	78	63.8 ± 1.9	1
NC800	Granodiorite porphyry	Biotite	7.69	7.0716×10^{-10}	76.2	62.9 ± 1.9	2
Camelback Mountain block (Camelback Mountain)							
NC806	Granodiorite	Biotite	8.46	5.8867×10^{-10}	60.5	42.5 ± 1.7	2
			6.56	3.7399×10^{-10}	50.9	39.2 ± 1.9	2
Cardigan Peak Block (east)							
L-844	Monzogranite (?)	Biotite	4.74	1.62×10^{-10}	21	19.6 ± 1.3	1
		Hornblende	0.394	1.38×10^{-11}	9	20.1 ± 3.5	1
NC44	Monzogranite	Biotite	9.17	2.874×10^{-10}	37.7	21.6 ± 0.9	2
Cardigan Peak block (west)							
NC807	Monzodiorite	Biotite	9.14	3.0445×10^{-10}	63.1	23.0 ± 0.7	2
		Amphibole	0.269	8.4234×10^{-10}	10.5	21.6 ± 1.8	2
Ajo Volcanics							
108	Andesite (?)	Whole rock				25.0 ± 2.7	3
81AM1767	Andesite	Biotite	8.23	2.861×10^{-10}	45	23.8 ± 0.8	4

References: 1, McDowell (1971) [ages recalculated after Dalrymple (1979)]; 2, this study; 3, Eberly and Stanley (1978); 4, Gray and Miller (1984)

Decay constants used in the age calculations are those recommended by Steiger and Jager (1977): $\lambda_{\epsilon+\epsilon'} = 0.581 \times 10^{-10}\ yr^{-1}$, $\lambda_\beta = 4.962 \times 10^{-10}\ yr^{-1}$, and $^{40}K/K_{tot} = 1.167 \times 10^{-4}$; the $\pm$ value represents the estimated analytical precision at one standard deviation and is based on experience with replicated analyses in the Menlo Park laboratories

This report presents the results of both paleomagnetic and geochronologic studies. The K-Ar geochronology was undertaken to test the hypothesis that the western pluton and the eastern mineralized intrusion are cogenetic. Because intrusive rocks lack conventional indicators of the paleohorizontal, paleomagnetic measurements were made on samples from plutonic and related igneous rocks of the Ajo mining district to estimte the structural disturbance these rocks have undergone since emplacement. Upon cooling below their magnetic blocking temperatures, igneous rocks acquire a remanent magnetization parallel to the ambient geomagnetic field that is stable over geologic time. Stable remanent magnetizations may also be acquired by rocks subject to subblocking-temperature chemical alteration. The direction of the geomagnetic field at Ajo for the time the plutonic and related igneous rocks cooled and became magnetic can be calculated from paleomagnetic directions for contemporaneous rocks at undisturbed sites elsewhere in North America. Thus, comparisons of paleomagnetic directions from igneous rocks in the Ajo district with the expected Late Cretaceous-early Tertiary and middle Tertiary geomagnetic-field directions provide initial postcooling horizontal datums for these rock units. The combination of geochronologic and paleomagnetic techniques is especially useful in constraining the timing of tectonic events and yields an increased understanding of the complex geologic history of this mining district.

Geologic Setting and K-Ar Geochronology

The Ajo mining district can be divided into two main structural blocks. To the east is the Camelback Mountain block which includes the New Cornelia mine, and to the west is the Cardigan Peak block which includes the main exposures of plutonic rocks in the district. These two blocks are divided by the eastward-dipping Gibson Arroyo fault which shows probable normal displacement (Gilluly, 1946) and which is in turn cut by the steeply dipping Little Ajo Mountain fault on the north side of the district (Fig. 1). The following brief geologic descriptions of these two structural blocks are after Gilluly (1946) and Cox and Ohta (1984) with appropriate changes in age and order of intrusion based on the unpublished U-Pb data and K-Ar data of this report. Because of their petrologic similarity, Gilluly (1946) applied the name Cornelia Quartz Monzonite to plutonic rocks in both structural blocks. The informal name Cornelia pluton has also been used to refer to these same rocks (Wadsworth, 1968). However, the new geochronologic data indicates that these plutonic rocks represent

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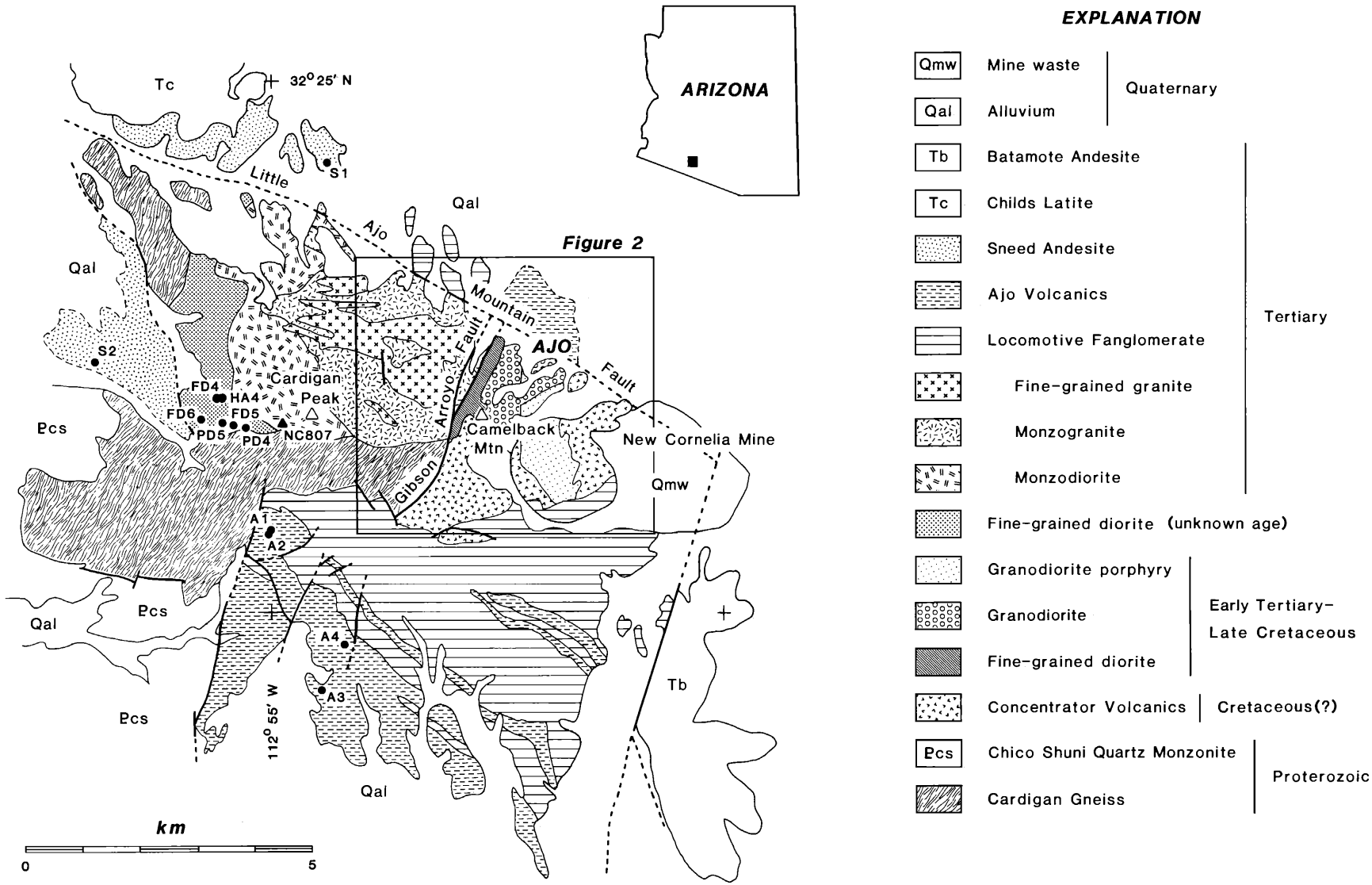


FIG. 1. Generalized geologic map of the Ajo mining district showing paleomagnetic (dots) and geochronologic (filled triangles) sampling sites. Geology is after Gilluly (1946) with modifications to the Cardigan Peak pluton according to Cox and Ohta (1984).

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two separate intrusions of differing ages. The names Cornelia Quartz Monzonite and Cornelia pluton are restricted in this report to the older Laramide plutonic rocks of the Camelback Mountain block and the informal name Cardigan Peak pluton is applied to the younger plutonic rocks of the Cardigan Peak block.

Camelback Mountain block

Deep drilling below the New Cornelia orebody has encountered the Proterozoic Cardigan Gneiss (Ronald Gibbs, oral commun., Phelps Dodge Corp., 1984), the basement and oldest rock of the district. Unconformably overlying the Cardigan Gneiss are the Cretaceous (?) Concentrator Volcanics. Consisting of a thick sequence (approximately 1 km) of intermediate to felsic flows, breccias, and lapilli tuffs, these rocks have been altered intensely by subsequent intrusive and hydrothermal activity. The Laramide plutonic rocks, here referred to as the Cornelia pluton, intrude the Concentrator Volcanics in the vicinity of the New Cornelia mine. A fine-grained diorite border facies of this pluton cuts the Cardigan Gneiss at the western edge of the block (Fig. 2). Exposed to the east of this diorite is quartz monzodiorite and medium- to coarse-grained granodiorite. Two different biotite fractions from a sample of the granodiorite from Camelback Mountain give K-Ar ages of 39.2 ± 1.9 and 42.5 ± 1.7 m.y. (Fig. 2 and Table 1). These ages are interpreted to be minimum ages caused by $^{40}Ar_{rad}$ loss due to thermal effects associated with the emplacement of the younger hornblende andesite dikes (see discussion). Gradational contacts separate the granodiorite from the granodiorite porphyry phase of the Cornelia pluton to the southeast which hosts the copper mineralization. The granodiorite porphyry has complex dikelike contacts with the Concentrator Volcanics, and it contains blocks of the Concentrator Volcanics and the fine-grained diorite. The alteration which includes the Cu mineralization is potassic (K-feldspar + biotite + anhydrite) with a chlorite overprint. Magnetite is present as veinlets and stable relict grains in the altered porphyry, a feature which contributes to the success of paleomagnetic studies of the orebody. A K-Ar age of 62.9 ± 1.9 m.y. for hydrothermal biotite in the New Cornelia mine corroborates the ages previously reported by McDowell

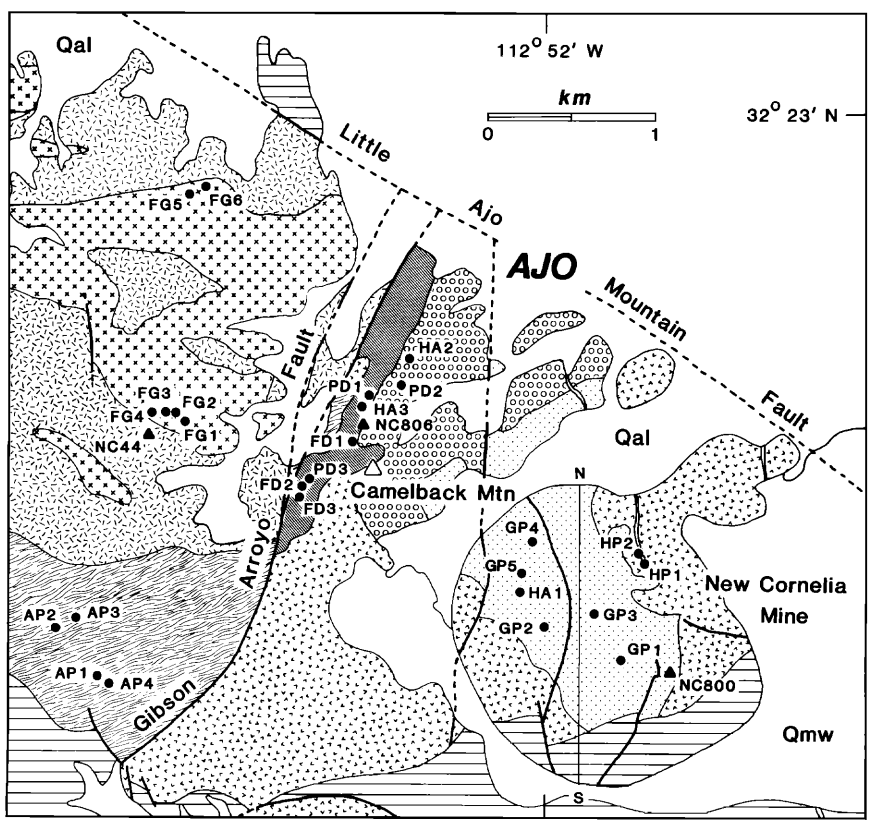


FIG. 2. Generalized geologic map of the Camelback Mountain block showing paleomagnetic (dots) and geochronologic (filled triangles) sampling sites. Units as in Figure 1. Geology of the New Cornelia mine is from an unpublished map of the Phelps Dodge Corporation. Line N-S through the mine indicates the trend of the cross section shown in Figure 10.

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(1971; Table 1). Feldspar porphyry dikes and hornblende andesite dikes intrude the Camelback Mountain block and cut rocks as young as the granodiorite porphyry. The feldspar porphyry dikes are altered in the mine and are probably of Laramide age. The hornblende andesite dikes generally cut the alteration and are likely of middle Tertiary age, but intrusive relationships with the granodiorite porphyry suggest that there is also an older generation of similar dikes roughly contemporaneous with the granodiorite porphyry.

The middle Tertiary Locomotive Fanglomerate rests on an irregular erosion surface in Laramide and older rocks. At least 2 km thick, the fanglomerate is interbedded with and capped by the Ajo Volcanics (Fig. 1) which consists of a thick sequence (≥1 km) of andesitic breccias, flows, and tuffs. Samples from the Ajo Volcanics have K-Ar biotite and whole-rock ages of 23.8 ± 0.8 and 25 ± 2.7 m.y. (Table 1), and the fanglomerate and volcanic rocks have been steeply tilted to the south 40° to 60°. Intruding the Locomotive Fanglomerate and the granodiorite porphyry of the Cornelia pluton are dikes of the Hospital Porphyry. These dikes are petrographically similar to and are probably contemporaneous with the Childs Latite which has been dated 18.3 ± 0.6 m.y. by K-Ar techniques at a site to the north of Ajo in the Crater Range (Gray and Miller, 1984).

Cardigan Peak block

The Proterozoic Cardigan Gneiss is widely exposed along the lower southern slopes of Cardigan peak and at the western margin of the Cardigan Peak block (Fig. 1). The Proterozoic Chico Shuni Quartz Monzonite (Gray et al., 1984) crops out farther to the southwest and is a dominant clast type in the Locomotive Fanglomerate west of the Gibson Arroyo fault. Only a small exposure of the Cretaceous (?) Concentrator Volcanics is found near the western edge of the block. These rocks are cut along the western margin of the Cardigan Peak block by a fine-grained diorite of uncertain age that is similar to the fine-grained diorite which crops out along the western margin of the Camelback Mountain block.

Forming the core of the Cardigan Peak block and the higher elevations of the Little Ajo Mountains area are middle Tertiary plutonic rocks here informally called the Cardigan Peak pluton (Fig. 1). Exposed in a broad zone inboard of the fine-grained diorite is monzodiorite which has reversely concordant hornblende and biotite K-Ar ages of 21.6 ± 1.8 and 23.0 ± 0.7 m.y., respectively (Table 1). Constituting the central part of the pluton is monzogranite with several textural variants. Although some of these rocks are similar in appearance to those which host the porphyry copper ore to the east, a sample from the eastern edge of the Cardigan peak pluton (Fig. 1) has a K-Ar age of 21.6 ± 0.9 m.y., nearly identical with those ages reported by McDowell (1971; Table 1). Within the monzogranite are one large and several small east-west-trending bodies of fine-grained granite. Abundant feldspar porphyry and hornblende andesite dikes, petrographically similiar to those of the Camelback Mountain block, intrude the fine-grained diorite and monzodiorite on the west side of the Cardigan Peak pluton. Some dikes extend from the monzodiorite generally a few tens of meters into the monzogranite but otherwise are absent from the core of the pluton. Numerous andesite porphyry dikes intrude the Cardigan Gneiss and are probably mid-Tertiary in age on the basis of paleomagnetic relations (see discussion). The Cardigan Peak pluton has been subjected to a sodic-calcic alteration around veinlets of epidote, hematite, and actinolite or, more rarely, tourmaline. This alteration is cut by the Gibson Arroyo fault. Hydrothermal effects beyond the periphery of the pluton on the east, south, and west appear to be younger than the Gibson Arroyo fault and consist of specularite-chalcopyrite veins that have been prospected for precious metals. These veins cut the Cardigan Gneiss and Concentrator Volcanics in the Cardigan mining district (near sites AP1-4, Fig. 2) and cut the Locomotive Fanglomerate and Ajo Volcanics at the Copper Ridge claims southeast of the New Cornelia mine waste dump. Specular hematite alteration in the granodiorite porphyry and Concentrator Volcanics in the New Cornelia mine (Gilluly, 1946, plate 27) is also related to this younger hydrothermal event.

Unconformably overlying the Cardigan Gneiss are the Locomotive Fanglomerate and Ajo Volcanics. Because of similar dips to the south (40°–60°), these units are considered continuous between the Camelback Mountain and Cardigan Peak blocks and appear to overlie the southern extension of the Gibson Arroyo fault. However, faults are difficult to map in the fanglomerate because of its crude bedding and the lack of distinctive marker beds, and faults with a similar strike but opposed dips to the Gibson Arroyo fault cut the Ajo Volcanics (Fig. 1). The Locomotive Fanglomerate also crops out north of the Little Ajo Mountain fault in and west of the town of Ajo. The Sneed Andesite overlies the fanglomerate to the north and the Chico Shuni Quartz Monzonite to the west of the Cardigan Peak block. A flow just above the Sneed Andesite in the Growler Mountains to the west of Ajo has been dated by K-Ar techniques at 22.0 ± 0.7 m.y., putting a minimum age on the Sneed Andesite (Gray and Miller, 1984).

Paleomagnetic Methods

Samples for paleomagnetic analysis were collected as 2.5-cm-diameter drill cores, using a portable ga-

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soline-powered drill, and were oriented in the field to an estimated accuracy of 2°, using magnetic and solar compasses and a clinometer. In general, samples were collected away from ridge crests to avoid outcrops contaminated by spurious lightning-induced magnetizations (Graham, 1961; Cox, 1962). Cylindrical specimens 2.2 cm in length were cut from the cores drilled in the field, and their remanent magnetization was measured with a cryogenic magnetometer. All samples were subjected to either progressive alternating field or thermal demagnetization. Alternating field demagnetization in peak fields between 1 and 100 millitesla (mT) was the primary technique used in this study and was carried out using a tumbling alternating field demagnetizer. Thermal demagnetizations were performed in air to temperatures in excess of 600°C employing a shielded furnace.

Components of magnetization were selected from demagnetization data using a least-squares method based on principal component analysis modified from Kirschvink (1980). Either lines (a single component), or for two sites, planes (two partially overlapping components) were fitted to the magnetization vector endpoints to obtain the components of remanence that were isolated by the demagnetization treatments. The statistical methods of Fisher (1953) were used in analyzing the averages of within-site line directions and the unit means. Poles to the fitted planes form an equatorial girdle about the common direction that corresponds to a Bingham distribution (Onstott, 1980), and for this reason Bingham statistics were used to calculate the mean direction for those sites (site-mean direction) exhibiting intersecting demagnetization planes. Structural corrections for the extrusive units were made by a simple rotation of the measured bedding plane to the horizontal about the strike axis. Dispersion in the declination of these corrected magnetic directions therefore could be introduced by unrecognized local plunge or rotations of structure (MacDonald, 1980).

Paleomagnetic Results

Camelback Mountain block

Within the open pit of the New Cornelia mine (Fig. 2), samples were collected at five sites in the granodiorite porphyry phase of the Cornelia pluton that hosts the Cu mineralization (sites GP1–5), at two sites in the Hospital Porphyry dike which cuts the granodiorite on the east wall of the pit (sites HP1 and 2), and at one site in a hornblende andesite dike on the west wall of the pit (site HA1).

Intensities of natural remanent magnetization average 0.2 ampere per meter (A m^{-1}) for the granodiorite porphyry and 2.1 A m^{-1} for the andesite dikes. Demagnetization of the dike rocks removed a single stable component of magnetization between 10 mT and 100 mT or 350° and 575°C (Fig. 3a), suggesting that titanomagnetite is the remanence carrier. For this stable magnetization, the site-mean directions (Fig. 4 and Table 2) group near the mid-Tertiary reference direction (Table 3) indicating that the dikes have not been significantly tilted since becoming magnetized.

Samples from the granodiorite porphyry, however, show more complex behavior upon demagnetization. These rocks tend to exhibit multicomponent magnetizations that are probably related to the emplacement and cooling of the granodiorite porphyry and to its remagnetization by reheating and/or chemical alteration associated with the emplacement of mid-Tertiary rocks in the district. Three components of magnetization are observed in samples from site GP1 and thermal demagnetization proved best in resolving all three components (Fig. 3b). An inconsistent component was removed in these rocks by 300°C (<15 mT), a second component was usually removed between 400° and 500°C (15–40 mT), and a third component was removed between 573° and 675°C (>100 mT). Although nearly obscured in some samples by the first and third components (Fig. 3b), the second component (GP1, Fig. 4) is considered the characteristic magnetization carried by titanomagnetite. This magnetization was probably acquired during initial cooling and potassic alteration of the granodiorite porphyry at approximately 64 m.y., because a later pervasive remagnetization most likely would have reset the K-Ar date of the porphyry as well. The high (>580°C) blocking temperatures for the third component indicate that it is carried by hematite; directions of the third component resolved with demagnetization lines from two samples are near the directions for the crosscutting dikes (Fig. 4). This suggests that the third component results from partial hematitization of the granodiorite at site GP1, perhaps associated with emplacement of the dikes. Two components of magnetization are observed in samples from site GP5. Upon demagnetization, a component with directions near those of the dikes (Fig. 4) was removed by 20 mT or 350°C, suggesting reheating and partial remagnetization at this site by the dikes. A second characteristic component was determined between 20 and 90 mT (500° and 570°C). Similar results for the characteristic magnetization were found in samples from site GP2, and the results for samples from site GP4 (Fig. 4) imply complete remagnetizaton by reheating of these rocks. A single characteristic component of magnetization was resolved between 15 and 100 mT (Fig. 3c) in samples from site GP3. The mean directions for a site of the characteristic component of magnetization for the granodiorite porphyry (Fig. 4 and Table 2) are far from the Late Cretaceous-early Tertiary reference direction for western North America (Table 3) and

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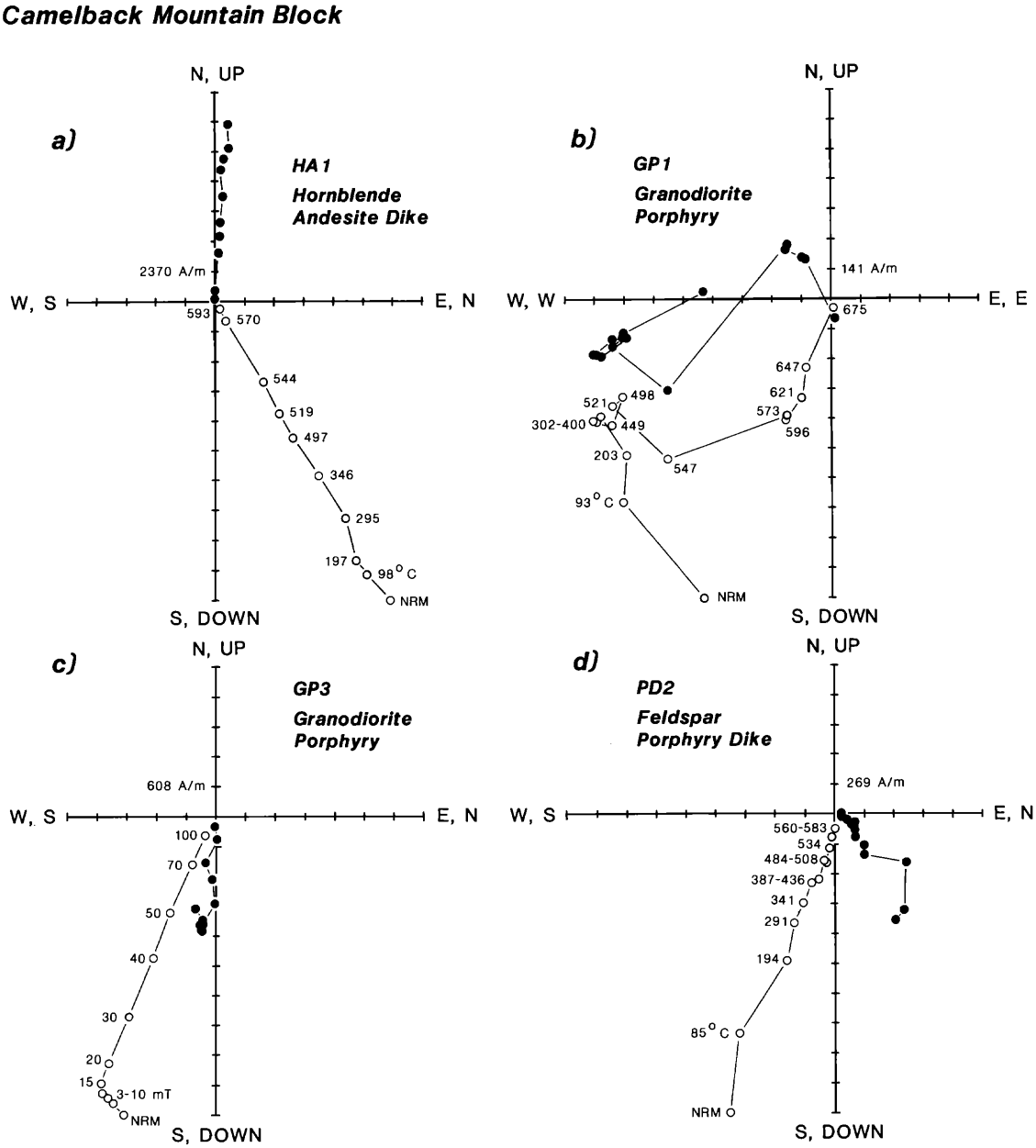


FIG. 3. Orthogonal projections of demagnetization vector endpoints for samples from the Camelback Mountain block. (a). Thermal demagnetization of a sample from a hornblende andesite dike in the New Cornelia mine showing a middle Tertiary component of magnetization. (b). Thermal demagnetization of a sample from the granodiorite porphyry in the mine showing multicomponent behavior (see text). (c). Alternating field demagnetization of a sample from the granodiorite porphyry in the mine showing a tilted Laramide component of magnetization. (d). Thermal demagnetization of a sample from a feldspar porphyry dike on Camelback Mountain showing a tilted Laramide magnetization. Filled symbols indicate projections of vector endpoints onto the horizontal plane and unfilled onto the vertical plane.

are considered to reflect the tectonic disturbance of the Camelback Mountain block since the initial cooling and magnetization of these rocks.

Sampling sites in the Camelback Mountain block are also located to the west of the New Cornelia mine on Camelback Mountain (Fig. 2). Samples were collected in hornblende andesite dikes at two sites (HA2 and 3), in feldspar porphyry dikes at three sites (PD1-3), and in the fine-grained diorite border phase of the Cornelia pluton at three sites (FD1-3). Although the

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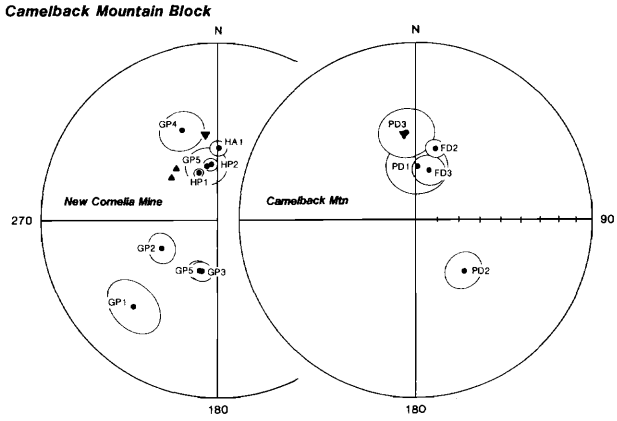


FIG. 4. Mean directions of sites uncorrected for tilt with 95 percent confidence limits for the Camelback Mountain block (Table 2). Large inverted triangle symbol indicates the predicted middle Tertiary geomagnetic field direction at Ajo calculated from an average pole for North America (Table 3). Small triangles indicate two remagnetized sample directions from site GP1 associated with hematite (see text). Diagrams are equal-area plots where filled symbols indicate projection from the lower hemisphere.

highest exposures on ridge crests were avoided, samples from three sites at higher elevations on Camelback Mountain (Table 2) have greater natural remanent magnetization intensities (by factors of 2 to 10) than similar lithologies at other sites. In addition, the directions for these samples after demagnetization are randomly dispersed (Watson, 1956), indicating that the rocks at these sites have been completely remagnetized by lightning strikes.

Natural remanent magnetization intensities for samples from the other sites in the Camelback Mountain block average 3.7 A m^{-1} for the feldspar porphyry dikes and 0.4 A m^{-1} for the fine-grained diorite. Several samples from sites PD1 and PD3 also have greater natural remanent magnetization intensities (by factors of 10 to 20) compared to other samples at the same site and highly scattered directions after demagnetization indicating that lightning currents have affected these sites as well. These dispersed directions have been discarded from the site means (Table 2) and an average natural remanent magnetization intensity for the feldspar porphyry dikes without these samples is 0.6 A m^{-1}. Alternating field (Fig. 3d) and thermal demagnetization of the remaining samples from Camelback Mountain generally removed spurious magnetizations below 15 mT (350°C) and a single component of magnetization by 100 mT (580°C), suggesting that titanomagnetite is the remanence carrier. One exception to single-component behavior is a sample from site PD3 that had a reversed polarity component of magnetization below 20 mT, and a normal polarity component between 40 and 90 mT. Most mean directions of a site for these components cluster near the middle Tertiary reference direction (Fig. 4) implying a middle Tertiary age for this magnetization. The mean direction for site PD2 is similar to that for the Laramide granodiorite in the New Cornelia mine suggesting an older age for this magnetization.

Cardigan Peak block

Sampling was limited to relatively unaltered fine-grained rocks within the Cardigan Peak block in order to obtain the best material with which to determine magnetizations associated with the initial emplacement and cooling of the rock units. Thus, samples were taken near its eastern margin (Fig. 2) in the fine-grained granite phase of the Cardigan Peak pluton at six sites (sites FG1–6), and in andesite porphyry dikes cutting the Cardigan Gneiss at four sites (sites AP1–4). Toward the western margin of the Cardigan Peak block (Fig. 1), samples were collected at three sites in fine-grained diorite (sites FD4–6), at one site in a hornblende andesite dike (site HA4), and at two sites in feldspar porphyry dikes (sites PD4 and 5).

Average natural remanent magnetization intensities range from 0.2 A m^{-1} for the feldspar porphyry dikes, and 0.4 A m^{-1} for the andesite porphyry dikes and fine-grained diorite, to 4.9 A m^{-1} for the fine-grained granite and hornblende andesite dike. Demagnetization of these rocks to <10 percent of their natural remanent magnetization intensities by 70 to 100 mT or 570° to 580°C indicates that titanomagnetite carries the stable remanence. All rock units from the Cardigan Peak block have both normal and reversed polarity components of magnetization. Sites in the andesite porphyry dikes tend to have samples with single component magnetizations consistently of one polarity or the other (Fig. 5a). However, a normal component was removed below 20 mT and a reversed component between 20 and 100 mT in samples from site AP3. Samples from the fine-grained diorite most clearly show both components. Normal polarity components were demagnetized by 30 mT or 500°C and reversed components were demagnetized between 30 and 100 mT or 500° and 570°C in the fine-grained diorite (Fig. 5b). Most samples from the other rock units exhibit demagnetization behavior suggesting the presence of both components, although least-squares lines usually can be fitted to only one (Fig. 5c). The reversed polarity component of magnetization is uniformly associated with higher coercivities and blocking temperatures. In a number of samples from the hornblende andesite dike, a single component was found upon demagnetization with spurious intermediate directions suggesting either the complete overlap of normal and reversed polarity components, preventing the use of demagnetization planes analysis, or the record of a transitional geomagnetic field. These directions are widely dispersed from the other site

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TABLE 2. Poles and Mean Paleomagnetic Directions of Sites for the Ajo Mining District

Site	I	D	$n(n_0)$	R	k	α_{95}	Plat	Plong
Camelback Mountain block (New Cornelia Mine)								
GP1	33.0	224.8	8(8)	7.6856	22	12.0	−23.9	200.0
GP2	61.5	243.8	7(7)	6.9249	80	6.8	5.1	205.6
GP3	65.8	196.8	6(6)	5.9756	205	4.7	−8.2	235.9
GP4	45.2	338.5	7(8)	6.8288	35	10.3	70.5	146.0
GP5	65.8	199.3	9(9)	8.9330	119	4.7	−7.8	234.3
	64.9	349.5	7(9)	6.8663	45	9.1	73.4	221.2
HP1	66.8	339.3	10(10)	9.9830	529	2.1	67.0	211.0
HP2	64.3	354.4	8(8)	7.9779	317	3.1	75.6	231.3
HA1	57.1	1.6	9(9)	8.9667	241	3.3	84.6	260.6
Camelback Mountain block (Camelback Mountain)								
HA2			11(11)	1.7098	1			
HA3			5(5)	2.6590	2			
FD1			6(6)	2.3751	1			
FD2	56.3	16.1	8(8)	7.9574	164	4.3	76.1	314.1
FD3	66.6	15.5	8(8)	7.8765	57	7.4	69.6	277.2
PD1	65.8	2.4	6(8)	5.8115	27	13.3	74.3	253.1
PD2	57.2	135.7	5(5)	4.9504	81	8.6	−8.6	281.0
PD3	49.6	354.0	4(8)	3.9429	53	12.8	84.5	137.6
Cardigan Peak block (east)								
FG1	54.4	15.0	3(8)	2.9827	115	11.5	77.3	321.4
FG2	61.8	2.8	5(8)	4.9517	83	8.5	79.2	257.9
FG3	49.5	344.8	9(9)	8.8551	55	7.0	76.9	152.3
FG4	−21.4	193.1	4(8)	3.8902	27	17.9	65.6	34.5
FG5	48.7	348.6	3(8)	2.9907	216	8.4	79.8	144.6
FG6	49.7	2.7	8(8)	7.8787	58	7.4	87.1	15.6
AP1	75.9	348.0	9(9)	8.8535	55	7.0	58.1	236.9
AP2	−37.6	178.7	7(8)	6.9918	731	2.2	78.7	73.4
AP3	−37.2	172.4	7(7)	6.9377	96	6.2	76.6	99.1
	55.7	2.0	6(7)	5.9839	311	3.8	85.8	269.5
AP4	76.9	23.7	9(9)	8.6466	23	11.1	54.3	264.1
Cardigan Peak block (west)								
FD4	−24.2	161.6	7(8)	6.7453	24	12.7	64.0	111.9
	61.0	13.5	7(8)	6.9711	208	4.2	75.6	291.1
FD5	−52.6	208.7	4(8)	3.9356	47	13.6	65.9	327.3
	67.6	354.1	8(8)	7.9627	188	4.1	71.3	235.3
FD6	−63.1	174.0	9(9)	8.8501	53	7.1	77.0	227.8
	69.4	359.2	9(9)	8.7285	29	9.6	69.3	245.7
PD4	70.0	322.9	9(9)	8.8851	70	6.2	56.0	207.6
PD5	55.0	9.5	8(8)	7.9321	103	5.5	81.5	312.7
HA4	−45.6	172.9	5(8)	4.8882	36	13.0	81.9	117.8
Volcanic rocks								
A1	−13.6	135.0	6(6)	5.9875	399	3.4	41.1	135.8
	−3.5	148.2 U					47.2	118.0
A2	−39.1	167.9 B	6(6)		(−13/−2)	(14/26)	75.2	116.5
	−14.1	2.8 U					50.4	62.7
A3	−51.7	160.2	9(9)	8.7142	28	9.9	73.3	162.6
	−25.3	188.7 U					69.4	42.5
A4	−38.2	163.0	8(8)	7.9464	131	4.9	71.4	125.6
	−13.7	180.8 U					64.6	65.2
S1	−25.7	215.9 B	8(8)		(−58/−2)	(6/9)	52.2	358.6
	−50.6	224.5 U					52.5	326.4
S2	−49.2	163.0	5(8)	4.9570	93	8.0	75.3	152.6
	−17.9	192.1 U					64.2	38.6

Notes: I, D, inclination and declination in degrees uncorrected for tilt (entries for volcanic rocks give both corrected and uncorrected (U) directions); $n(n_0)$, number of samples averaged (number of samples measured); R, vector sum; k, precision parameter (Fisher, 1953); α_{95}, radius of 95% confidence in degrees; Plat, Plong, latitude and east longitude of virtual geomagnetic poles in degrees. Site prefixes: GP, granodiorite porphyry; HP, Hospital Porphyry; HA, hornblende andesite dike; FD, fine-grained diorite; PD, feldspar porphyry dike; FG, fine-grained granite; AP, andesite porphyry dike; A, Ajo Volcanics; S, Sneed Andesite. B = directions resolved with demagnetization planes and Bingham statistics (Onstott, 1980); for α_{95} and k the two Bingham 95% confidence limits and two Bingham concentration parameters are shown, respectively

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TABLE 3. Mean and Reference Paleomagnetic Directions and Pole Positions for the Ajo Mining District

	I	D	n	R	k	α_{95}	Plat	Plong
Middle Tertiary								
Observed	53.8	356.0	35	33.1906	19	5.7	86.1	188.7
Reference	50.3	352.2				3.6/3.7	83.2	148.0
(Flattening of observed inclination at site = −3.5 ± 5.6)								
(Rotation of site = 3.8 ± 8.4)								
Laramide								
Observed	62.1	204.3	5	4.6269	11	24.5	−11.1	229.3
Corrected	7.7	197.3					−50.2	219.5
Reference	−56.4	171.3				2.4/3.1	−81.5	12.6
(Flattening of corrected inclination at site = 64.1 ± 19.2)								
(Rotation of site = 25.9 ± 19.5)								

Notes: I, D, inclination and declination in degrees; n, number of sites averaged; R, vector sum; k, precision parameter (Fisher, 1953); α_{95}, radius of 95% confidence in degrees (two values indicate 95% confidence of I and D, respectively, calculated from A_{95} of reference pole); Plat, Plong, latitude and east longitude of virtual geomagnetic poles in degrees. North American reference poles are from Diehl et al. (1983). Corrected Laramide direction: corrected for tilt of Locomotive Fanglomerate (strike = 101°/dip = 55°; average of 37 measurements near the southern rim of the New Cornelia mine), see text. Flattening and rotation of mean direction with respect to reference direction, in degrees with 95% confidence limits, are calculated after Beck (1980) as modified by Demarest (1983)

means and have been excluded from the site-mean calculations (Table 2).

The results for the fine-grained granite are more problematical. Most of the samples from most of the sites in this unit have components of magnetization that are similar to those observed in the other units

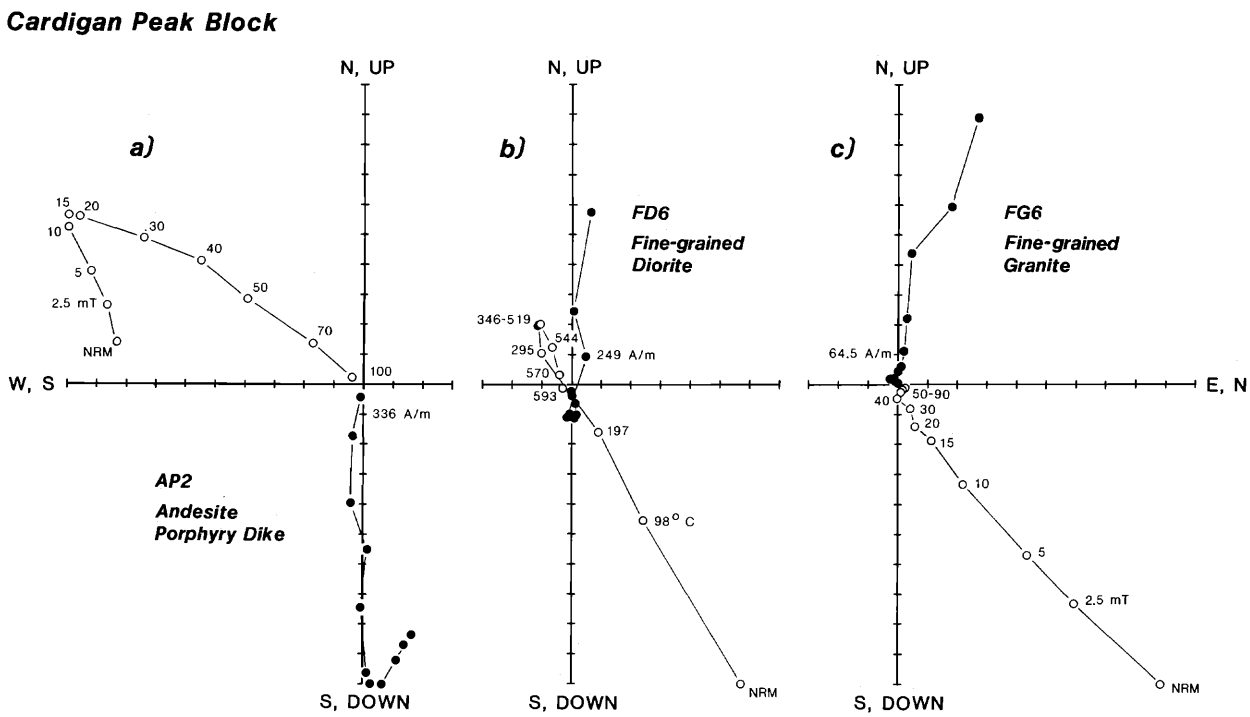


FIG. 5. Orthogonal projections of demagnetization vector endpoints for samples from the Cardigan Peak block. (a). Alternating field demagnetization of a sample from an andesite porphyry dike that cuts the Cardigan Gneiss. (b). Thermal demagnetization of a sample from the fine-grained diorite near the western margin of the Cardigan Peak block. (c). Alternating field demagnetization of a sample from the fine-grained granite near the eastern margin of the block. Filled symbols indicate projections of vector endpoints onto the horizontal plane and unfilled onto the vertical plane.

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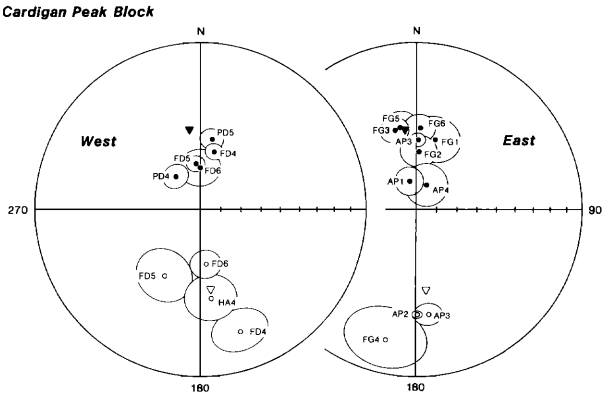


FIG. 6. Mean directions of sites uncorrected for tilt for the Cardigan Peak block (Table 2). Triangle symbols indicate the predicted middle Tertiary geomagnetic field direction at Ajo (Table 3). Filled symbols indicate projections from the lower hemisphere and unfilled from the upper hemisphere.

of the Cardigan Peak block (Fig. 6 and Table 2). These mean directions cluster near the middle Tertiary geomagnetic field direction for western North America and suggest that the Cardigan Peak block has not been significantly tilted since these rocks were magnetized. However, five samples from site FG1, one sample from sites FG2 and FG4, and two samples from site FG5 have single-component magnetizations with an average direction (inclination = 40.5°, declination = 246.6°, n = 9, precision parameter = 40, radius of 95% confidence in degrees = 8.2°) that appears to have undergone tilting similar to that affecting the Laramide units from the Camelback Mountain block. Eight other samples from sites FG2, FG4, and FG5 show dispersed directions, with easterly and southerly declinations and shallow inclinations, that may represent spurious resultants of superimposed components. These directions also have been discarded from the site-mean direction calculations for a site (Table 2).

Volcanic rocks

The Ajo Volcanics were sampled at four sites (A1–4) to the southwest of Ajo and the Sneed Andesite was sampled at two sites (S1 and 2) to the northwest and west of Ajo (Fig. 1). These rocks of known attitude were sampled to confirm the direction of the middle Tertiary geomagnetic field. Attitudes measured on flow layering within the Sneed Andesite and on interbedded sediment layers within the Ajo Volcanics are used as structural corrections for these rocks.

Natural remanent magnetization intensities generally average around 0.5 A m^{-1} for the volcanic rocks, although samples from site S1 have an average intensity of 12.1 A m^{-1}. Demagnetization planes are needed to resolve the direction of stable remanence for this site, suggesting that at this locality the andesite was partially remagnetized by lightning currents. Thermal demagnetization of most samples from the Ajo Volcanics was necessary because hematite is the dominant stable remanence carrier in these rocks. Usually, a single reversed polarity component of magnetization was removed from these rocks between 500° and 675°C (Fig. 7a). However, both normal and reversed polarity components are observed in samples from site A2; partial overlap of the two components requires the use of demagnetization planes to recover the stable direction of magnetization for this site (Table 2). Successful demagnetization of samples by 100 mT or 580°C points to titanomagnetite as the remanence carrier in the Sneed Andesite (Fig. 7b). Normal (generally <10 mT or 300°C) and reversed polarity components also occur in samples from site S2, and a transitional geomagnetic field could explain the widely divergent directions in three samples from this site. Mean directions of a site for the volcanic rocks before and after structural correction are plotted in Figure 8. The directions are near the predicted field direction after structural correction, excepting sites A1 and S1. The divergent declinations of these sites could be due either to secular variation of the geomagnetic field or to local rotations not accounted for in the structural correction.

Discussion and Conclusions

The K-Ar ages presented in this study apparently corroborate previously published K-Ar data (McDowell, 1971) and unpublished U-Pb data (J. S. Stacey and J. L. Wooden, written commun., 1986; R. M., Tosdal, written commun., 1987) in suggesting a significant temporal separation between the Laramide Cornelia pluton with its associated porphyry copper mineralization and the middle Tertiary Cardigan Peak pluton (Table 1). The K-Ar ages also seem to imply that the Cardigan Peak pluton is closely related in time to the eruption of the Ajo Volcanics and Sneed Andesite. However, paleomagnetic evidence indicates that the remanent magnetization of the Cardigan Peak pluton has been reset by the intrusion of dikes younger than the Locomotive Fanglomerate and Ajo Volcanics, and this younger intrusive event may have reset the K-Ar age of the pluton as well.

The paleomagnetic results of this study, in conjunction with the available geochronologic data, provide for a revised structural history of the Ajo mining district. Correction for the tilt of the Locomotive Fanglomerate (101°/55°) moves the observed average paleomagnetic direction for the Laramide rocks of the Camelback Mountain block to a shallow inclination with southerly declinations (Fig. 9). Another 64° (±19°) of southward tilt and 26° (±20°) of clockwise rotation must be removed to bring the Laramide direction into coincidence with the Late Cretaceous-

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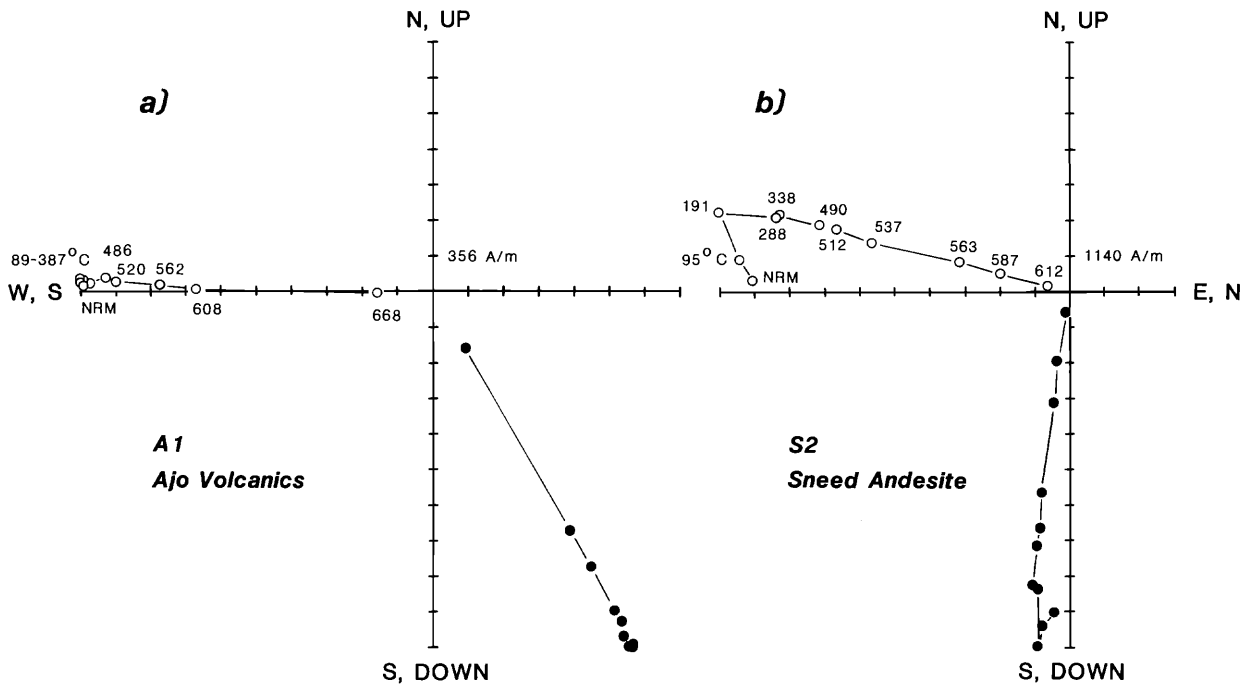


FIG. 7. Orthogonal projections of demagnetization vector endpoints for samples from the volcanic rocks. (a). Thermal demagnetization of a sample from the Ajo Volcanics showing that hematite is the dominant remanence carrier in these rocks. (b). Thermal demagnetization of a sample from the Sneed Andesite showing that titanomagnetite is the dominant remanence carrier in these rocks. Filled symbols indicate projections of vector endpoints onto the horizontal plane and unfilled onto the vertical plane.

early Tertiary reference direction at Ajo (Table 3). This motion could also be accomplished by removing a single rotation of 68° about an approximate strike of 105° which brings the complex dikelike intrusive contacts of the intrusion to near vertical (Fig. 10). This near-vertical dike swarm is similar to that observed in the upper levels of the quartz monzonite porphyry intrusion associated with the Dos Pobres orebody of the Lone Star mining district (fig. 16.5/ Langton and Williams, 1982). Thus, it appears that the ore-bearing granodiorite porphyry at Ajo was initially magnetized in a reversed polarity geomagnetic field and tilted 68° to the south before, and another 55° south after, deposition of the Locomotive Fanglomerate and Ajo Volcanics.

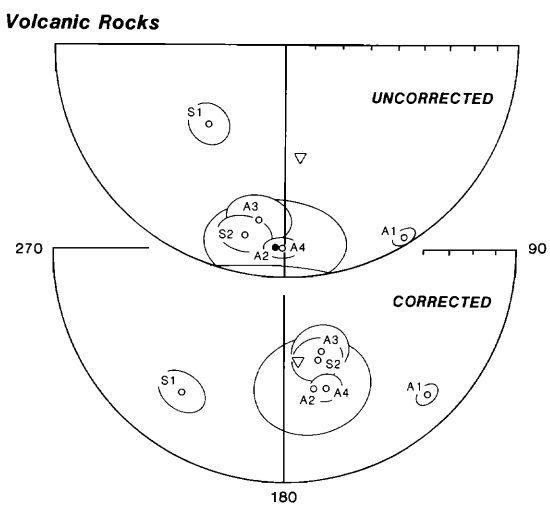


FIG. 8. Mean directions of sites for the volcanic rocks uncorrected and corrected for structural tilt (Table 2). Triangle symbols indicate the predicted middle Tertiary geomagnetic field direction at Ajo (Table 3). Filled symbols indicate projections from the lower hemisphere and unfilled from the upper hemisphere.

The untilted paleomagnetic directions for the feldspar porphyry and hornblende andesite dikes sampled in the Cardigan Peak block suggest that they are younger than the tilting of the Locomotive Fanglomerate and Ajo Volcanics. Gilluly (1946) had considered these feldspar porphyry dikes to be Laramide in age and contemporaneous with those of the Camelback Mountain block and had recognized at least two and possibly three episodes of hornblende andesite dikes. Indistinguishable in appearance from dikes of the older groups, members of the youngest group of hornblende andesite dikes clearly postdate the Locomotive Fanglomerate. Emplacement of the younger feldspar porphyry and hornblende andesite dikes

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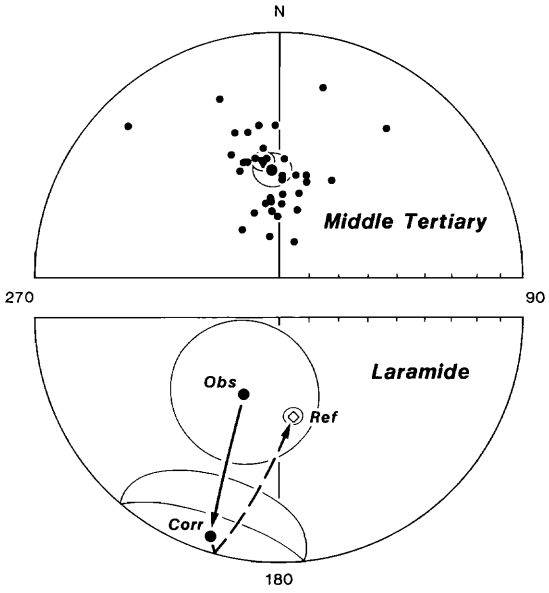


FIG. 9. Upper net shows the mean directions (dots) of sites for units considered middle Tertiary in age, their average direction (large dot) with 95 percent confidence limits, and the middle Tertiary reference direction at Ajo (inverted triangle) with 95 percent confidence limits (Table 3). Lower net shows an average of the Laramide directions observed in rocks from the Camelback Mountain block (Obs), this average direction corrected for the tilt of the Locomotive Fanglomerate (Corr), and the Late Cretaceous-early Tertiary reference direction at Ajo (Ref), all with 95 percent confidence limits (Table 3).

probably caused reheating and remagnetization of the enclosing rock and could explain the untilted paleomagnetic directions for the samples from the Cardigan Peak pluton. The feldspar porphyry dikes, with subordinate hornblende andesite dikes, are numerous near the western margin of the Cardigan Peak block, and probably have completely remagnetized the fine-grained diorite host rock. Although dikes do not cut the fine-grained granite, its untilted component of magnetization is inferred to be a partial overprint caused by heating and alteration associated with emplacement of the dikes. The almost obliterated primary (?) component for the fine-grained granite is tentatively interpreted as a tilted reversed polarity middle Tertiary magnetization. This would imply that the pre-Locomotive Fanglomerate southward tilt of $\simeq 70°$, which affected the Laramide units to the east, occurred after emplacement and initial magnetization of the Cardigan Peak pluton.

The post-tilting episode of dike emplacement and remagnetization appears to have affected rocks in the Camelback Mountain block as well. A feldspar porphyry dike on Camelback Mountain (PD2) with an older tilted magnetization intrudes rocks with post-tilting magnetizations at other localities that are probably remagnetized (FD2 and 3, Fig. 4). Feldspar porphyry dikes of similar appearance sampled elsewhere on Camelback Mountain (PD1 and 3) are probably remagnetized as well. The partially reset K-Ar age of a granodiorite sample from Camelback Mountain (Table 1) is consistent with an event of younger reheating and remagnetization in this vicinity because of the approximate equivalence of magnetic and isotopic blocking temperatures. Samples within the mine to the east also appear to have been remagnetized either by reheating (GP4 and 5) or by hydrothermal alteration and hematitization (GP1) associated with the younger dike emplacement. All untilted mean directions of a site for rocks in the Ajo mining district that are inferred to be a middle Tertiary age have been plotted in Figure 9. An average of these directions is not significantly different from the predicted direction at Ajo for this time (Table 3). These data indicate that significant tectonic activity in the Ajo district had ceased by the time the younger dikes cooled and were magnetized (approximately 20 m.y.) and they do not support the $14° \pm 12°$ counterclockwise rotation of southwestern Arizona relative to cratonic North America since the late Miocene reported by Calderone and Butler (1984).

The timing of movement on the Gibson Arroyo fault becomes a problem in light of the close similarity in K-Ar ages of the Cardigan Peak pluton and the Ajo Volcanics. Considering the rough equivalence of magnetic and K-Ar isotopic blocking temperatures, the inferred remagnetization of the Cardigan Peak pluton suggests that its K-Ar date was also reset by intrusion of the younger dikes, although the preliminary U-Pb data indicate that the Cardigan Peak pluton is no older than 30 m.y. (R. M. Tosdal, written commun., 1987). Displacement along the Gibson Arroyo

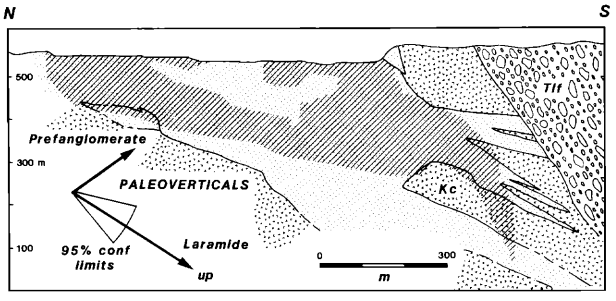


FIG. 10. Cross section through the New Cornelia mine (after Gilluly, 1946) depicting the granodiorite porphyry (stippled pattern) which intrudes the Concentrator Volcanics (Kc). These units are unconformably overlain by the tilted Locomotive Fanglomerate (Tlf). Diagonally ruled area indicates zone of >0.5 percent Cu. Paleovertical for the fanglomerate is calculated from an average of 37 attitude measurements made in the fanglomerate near the southern rim of the New Cornelia mine; the Laramide paleovertical is calculated from the paleomagnetic data (Table 3 and Fig. 9). The trend of this cross section is shown as line N-S in Figure 2.

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fault is younger than the sodic-calcic alteration which affects the Cardigan Peak pluton and is great enough to remove all indication of this alteration from the eastern side of the fault. The best evidence available indicates that this movement predates the deposition of the Locomotive Fanglomerate and overlying Ajo Volcanics, but the untilted hornblende andesite dikes in the Cardigan Gneiss appear to have undergone left-lateral drag near the fault and hornblende andesite dikes in the Camelback Mountain block are cut by the fault (Gilluly, 1946). These relations suggest that a second episode of movement occurred along the Gibson Arroyo fault after tilting of the Locomotive Fanglomerate which was insufficient to disrupt this unit.

In summary, the paleomagnetic and geochronologic data discussed in this report indicate the following sequence of middle Tertiary events in the Ajo mining district: (1) emplacement of the Cardigan Peak pluton and solidification, fracturing, and sodic-calcic alteration of the rock unit around 25 to 30 m.y.; (2) movement along the Gibson Arroyo fault; (3) unroofing of the pluton to within a kilometer of its upper contact, perhaps associated with the $\simeq 70°$ southward tilt which affected the paleomagnetic directions of the Laramide rocks in the district; (4) syntectonic deposition of the Locomotive Fanglomerate and Ajo Volcanics at 24 to 25 m.y., possibly concurrent with some southward tilting; (5) uplift and further tilting to the south (totaling 40°–60°) along the Little Ajo Mountain fault; (6) emplacement of the youngest feldspar porphyry and hornblende andesite dikes at about 20 to 23 m.y.; and (7) minor (?) oblique movement along the Gibson Arroyo fault.

Acknowledgments

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May 8, December 9, 1986

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